Can Frozen Text and Code Embeddings Screen Coding-Agent Failures?

Tun-Chieh Yang (315553010) **Jia-Cheng Wu (314554059)** **Min-Quan Wang (315553008)**
Wei-Chia Chang (315553011) Chun-Ting Hou (315554022)
Team G04, Data Mining (Fall 2026), NYCU
https://github.com/calculusfkyou/nycu-data-mining-assignment-2-115-1

Abstract

Release engineers must screen completed coding-agent runs before running expensive integration test suites or merging candidate diffs into production repositories. We evaluate task-end failure screening (**Category 4: Text / Logs**) across 284 official `SWE-agent` (`GPT-4`) trajectories (54 normal, 230 failure; 80.99% prevalence) using local, frozen `all-MiniLM-L6-v2` log and `codebert-base` diff embeddings, validated once on an untouched holdout of 271 `Claude-3-Opus` runs (35 normal, 236 failure). Under 5-fold stratified evaluation, primary Tier 2 dual-encoder k-NN achieves threshold-free Average Precision Fold AP $= 0.841 \pm 0.02$ and pooled BA $= 0.481$ (0.479 ± 0.05) under an operational validation FPR $\leq 20\%$ review-capacity cap, flagging $14/54$ successes (FPR $= 0.259$) while missing $179/230$ failures (recall $= 0.222$; vs. 0.957 under min-cost thresholding). Primary `T2` trails Tier 1 rule heuristics (BA $= 0.526$) and Tier 3 supervised logistic regression (BA $= 0.596$, $p = 0.003$). Single-factor ablations explain why equal-weight fusion fails: noisy `MiniLM` log distances exhibit $56\times$ larger variance than `CodeBERT` patch distances, contributing 98.24% of distance variance and drowning out the patch channel (`CodeBERT` patch-only reaches 5-fold mean BA $= 0.595 \pm 0.04$, pooled BA $= 0.594$, ΔBA $= +0.113$, $p = 0.012$). Parameter-free `Inductive ECOD` on 23 clean numeric features reaches BA $= 0.657$ (FPR $= 0.130$, $p = 0.004$) and `Conditional Residual` reaches BA $= 0.675$ ($p = 0.011$). The untouched `Claude-3-Opus` holdout directionally preserves rankings (`T2`: $0.542/0.536$, `CodeBERT`: $0.620/0.592$, `ECOD`: $0.645/0.645$, `Conditional Residual`: $0.672/0.708$, `T3`: $0.691/0.711$). Under our stakeholder cost model ($1,200/escaped failure vs. $30/review; break-even $\pi^* = 2.5\%$), universal review ($15,000) dominates automated gating at 81% prevalence, whereas anomaly detectors uniquely catch 47.1% ($8/17$) of rare 6.0% ($17/284$) *silent semantic failures* where surface rules catch $0/17$ (0.0%).

1 Introduction

Autonomous coding agents iteratively issue shell commands, inspect tracebacks, and edit files to resolve software issues (Yang et al., 2024), but frequently fail regression suites (Jimenez et al., 2024). Our stakeholder is a **release engineer** triaging completed runs before running costly full-repository integration suites: an alert routes a patch to mandatory human review, whereas unflagged runs proceed to standard CI testing (a 40-minute suite costing ∼$150 compute/run, never autonomous release). We formulate screening at **task-end** rather than prefix stopping because candidate diffs exist only at completion, and prefix aborts prematurely kill recoverable debugging loops (e.g., resolved `scikit-learn-14894` executes 25 steps with 19 error observations before fixing the bug).

Our study falls under **Category 4: Text / Logs**. Agent logs mix shell commands, tracebacks, and unified git diffs. Scalar counts discard code semantics, yet pre-trained encoders measure topical similarity rather than functional correctness. We ask: *Can local, frozen text and code embeddings improve task-end failure screening over clean numeric workflow statistics under a shared normal-reference protocol and false-alarm constraint?*

We model resolved runs ($y = 0$, normal reference) as baseline and failures ($y = 1$) as targets. Surface counts poorly separate outcomes: resolved `django-13230` takes 15 steps with 9 error observations and is falsely flagged by our primary detector ($s = 0.1128 > \tau = 0.1087$), whereas failed `sympy-12454` finishes in 11 steps with 3 cumulative error observations (0 on final step) and escapes ($s = 0.0248 < \tau = 0.0731$). Crucially, failures constitute 80.99% ($230/284$) of runs, so failure triage is not a classic rare-event setting ($\pi \gg 10\%$). We therefore formulate two operational regimes: (1) **High-throughput failure triage**: screening common crashes to avoid wasting $150 CI suites, where minority successes (19.01%, $54/284$) define a coherent normal manifold; and (2) **Rare silent se-**

mantic failure detection: catching clean-exiting patches (last_step_error $= 0$, error_count ≤ 3, non-empty diff) that silently fail regression tests (6.0%, 17/284), a true rare-event anomaly setting ($\pi = 6.0\% < 10\%$) where surface rules are blind (0/17 caught, 0.0%) and anomaly detectors provide unique value (47.1%). Contributions: (1) shared-split 3-tier benchmark across four representations (`R0`–`R3`) with paired t-tests, bootstrap CIs, and latencies (Table 2); (2) variance and neighbor proofs closing three hypotheses (Tables 3–4); (3) once-only confirmatory holdout on 271 untouched `Claude-3-Opus` runs; and (4) cost-aligned threshold calibration ($\pi^* = 2.5\%$, Table 5) and step/feature attributions (Table 6, Figure 5).

2 Related Work

`SWE-bench` (Jimenez et al., 2024) evaluates patch generation across 12 Python repositories (1.96% resolved; 4.33% on 300-task `Lite`). `SWE-agent` (Yang et al., 2024) achieves 12.5% pass@1 on full `SWE-bench` (18.0% on `Lite`). Unlike prior work evaluating patches via execution, we study **Category 4: Text / Logs** pre-execution screening. Sentence-BERT (Reimers and Gurevych, 2019) learns Siamese transformer embeddings; we use frozen `all-MiniLM-L6-v2` (384-D) with chunk pooling. `CodeBERT` (Feng et al., 2020) pre-trains bimodal RoBERTa (768-D); we adapt frozen `codebert-base` to embed diff hunks without fine-tuning. Among anomaly detectors, Deep SVDD (Ruff et al., 2018) risks hypersphere collapse when $n_{\text{ref}} \approx 32$. We instead pair non-parametric cosine k-NN (Tier 2) with tabular `Isolation Forest` (Liu et al., 2008) and an **inductive** adaptation of parameter-free `ECOD` (Li et al., 2022) freezing empirical tail CDFs on normal training folds, alongside Tier 3 supervised logistic regression and `XGBoost` (Chen and Guestrin, 2016).

3 Task Definition and Data

Primary cohort and untouched confirmatory holdout. Our primary dataset comprises all $N = 284$ generated trajectories from official `20240402_sweagent_gpt4` `SWE-bench Lite` runs[1] (16 `no_generation` aborts excluded a priori): $n_0 = 54$ resolved ($y = 0$, normal reference, 19.01%) and $n_1 = 230$ unresolved ($y = 1$, failure, 80.99%). Outer 5-fold stratification is by binary label $y \in \{0, 1\}$ (seed 42); repo stratification was unnecessary as each fold spans ≥ 10 distinct repositories (Fold 0 has 11). Because `gpt4` was inspected during exploratory debugging, confirmatory validity requires an untouched cohort: all $N = 271$ generated runs from `20240402_sweagent_claude3opus` (29 `no_generation` excluded; $n_0 = 35$ normal, 12.92%; $n_1 = 236$ failure, 87.08%). Provenance and labels are archived in `Dataset/provenance.json` and `Dataset/v2/claude3opus_provenance.json`. We retain natural prevalence (80.99%, baseline AP $= 0.810$) and elevate the rare 6.0% (17/284) **silent semantic failure** subtype (`last_step_error=0`, error_count ≤ 3, non-empty diff, $y = 1$) to a first-class target.

Leakage hygiene and representations (`R0`–`R3`). Feature extraction uses only `action`, `observation`, and `submission` diffs; `thought`/`response` text, gold patches, evaluator logs, exit codes, and instance/repo IDs are stripped. Normalization removes ANSI codes, hex addresses, and `<WORKDIR>` prefixes while preserving paths and tracebacks (verified via hash audits and 20 inspections). Four representations: (1) **`R0` / `R0-clean` (24 / 23 features):** 16 workflow counts (steps, errors, streaks, stalls, tokens, cost) and 8 patch metrics (chars, lines, files, hunks, edit ratio, test flag). `R0-clean` drops duplicate `total_commands` ($\rho = 1.000$), applies $\log(1 + x)$ to 16 skewed counts, keeps 4 binary flags, and robust-scales via median/IQR fit *strictly on* $y = 0$ *training rows*; (2) **`R1`:** char 3–5-gram TF-IDF ($\leq$ 20,000 features/channel); (3) **`R2`:** 384-D `MiniLM` log embeddings; and (4) **`R3`:** dual 384-D `MiniLM` log (l_x) + 768-D `CodeBERT` diff (p_x) embeddings.

Feature	Norm. (IQR)	Fail. (IQR)	W	D_{KS}	p-val
Received tokens	1377 (1042)	2934 (2658)	0.999	0.520	5.5e-07
API cost (USD)	1.24 (0.91)	2.56 (2.64)	0.892	0.440	4.5e-05
Trajectory steps	12 (8)	24 (18)	0.885	0.426	9.1e-05
Error observations	3 (3)	7 (10)	0.709	0.408	2.1e-04
Patch chars	822 (500)	1372 (1321)	0.203	0.385	5.7e-04
Patch lines added	4 (6)	17 (28)	0.178	0.554	6.3e-08

Table 1: Fold-0 training-only ($n = 170$: 32 normal, 138 failure) medians (IQR), standardized Wasserstein distance W, and two-sample KS statistics (D_{KS}, p-value).

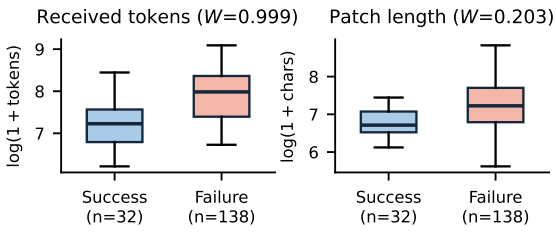


Figure 1: Fold-0 training-only $\log(1 + x)$ boxplots for received tokens ($W = 0.999$) and patch length ($W = 0.203$).

4 Exploratory Feature Analysis

On Fold-0 training data ($n = 170$: 32 normal, 138 failure), Table 1 and Figure 1 show failures

[1] https://github.com/SWE-bench/experiments/tree/main/evaluation/lite/20240402_sweagent_gpt4

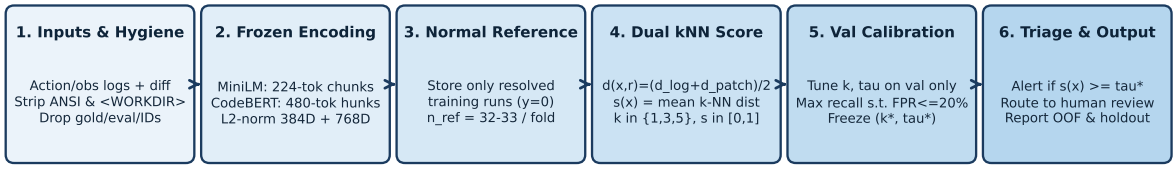


Figure 2: Six-checkpoint end-to-end architecture for our primary Tier 2 normal-reference dual-encoder k-NN pipeline (`knn_R3_T2`). All encoding, reference indexing, and threshold calibration occur strictly inside each fold.

consume $\sim 2\times$ more received tokens (2,934 vs. 1,377, $W = 0.999$, $p = 5.5 \times 10^{-7}$) and steps (24 vs. 12, $W = 0.885$) and produce larger diffs (17 vs. 4 lines added, $D_{\text{KS}} = 0.554$, $p = 6.3 \times 10^{-8}$), but show heavy tails, quartile overlap, and collinearity ($\rho = 0.985$ tokens/cost; $\rho = 0.941$ steps/observations). This motivates three falsifiable hypotheses: **H1 (Representation):** dual frozen embeddings (`R3`) improve k-NN BA over `R0-clean` and `R1` TF-IDF; **H2 (Channel attribution):** `CodeBERT` diff embeddings add discriminative signal beyond `MiniLM` logs (`R2`); and **H3 (AD contamination):** replacing 3.1% (1/32) and 9.4% (3/32) of normal references with failures degrades `T2` and `Inductive ECOD` BA.

5 Method

Six-checkpoint primary pipeline (Tier 2: `knn_R3_T2`). Figure 2 depicts our pipeline: (1) **Inputs & Hygiene** normalizes permitted logs and diffs; (2) **Frozen Encoding** chunks steps into 224-token windows with 32-token overlap (`all-MiniLM-L6-v2`), mean-pools non-special tokens, and ℓ_2-normalizes to $l_x \in \mathbb{R}^{384}$; unified diffs are chunked into 480-token windows with 64-token overlap (`codebert-base`), token-weight pooled, and ℓ_2-normalized to $p_x \in \mathbb{R}^{768}$; (3) **Normal Reference Index** stores only resolved training runs $\mathcal{R}_{\text{norm}}$ ($y = 0$, $n_{\text{ref}} \in \{32, 33\}$); (4) **Dual k-NN Score** computes cosine distances $d_{\log}(x, r) = \frac{1-\cos(l_x, l_r)}{2}$ and $d_{\text{patch}}(x, r) = \frac{1-\cos(p_x, p_r)}{2}$ (0 if both empty, 1 if one empty), fusing via unweighted average

$$d(x, r) = \tfrac{1}{2}\big(d_{\log}(x, r) + d_{\text{patch}}(x, r)\big) \in [0, 1], \quad (1)$$

and scores $s(x) = \frac{1}{k}\sum_{r \in \mathcal{N}_k(x; \mathcal{R}_{\text{norm}})} d(x, r)$ for $k \in \{1, 3, 5\}$ ($\leq 15.6\%$ of $n_{\text{ref}} = 32$). Equal weighting ($w_{\log} = w_{\text{patch}} = 0.5$) in Eq. (1) represents an uninformative maximum-entropy prior: terminal logs and diffs appear equally salient a priori, before variance decomposition reveals channel imbalance; (5) **Validation Calibration** selects (k^*, τ^*) maximizing recall subject to validation FPR $\leq 20\%$, a review-capacity SLO preventing triage alert fatigue (vs. min-cost F_2 in Ablation 3); and (6) **Triage & Output** freezes (k^*, τ^*) to flag test runs with $s(x) \geq \tau^*$.

Baselines and sample-size hyperparameter justification. Hyperparameters are bounded by $n_{\text{ref}} \approx 32$ normal fitting rows ($n_{\text{fit}} \approx 170$ total): **Tier 1 (`heuristic_T1`)** scores $\text{error_rate} + 0.1\,\text{repeated_tool_streak} + \mathbb{I}[\text{last_step_error}] + 0.5\,(1 - \mathbb{I}[\text{repro_command}])$, following CI priorities where terminal crashes incur unit penalty (1.0), while tool looping (0.1) and missing repro commands (0.5) carry fractional penalties. **Tier 2 Tabular Normal Controls:** `Inductive ECOD` (Li et al., 2022) has *zero* tuned hyperparameters, freezing smoothed tail CDFs $(1 + \#\{r_j \leq x_j\})/(n_{\text{ref}} + 1)$ on $\mathcal{R}_{\text{norm}}$ across all 23 `R0-clean` features. `Isolation Forest` (Liu et al., 2008) uses 200 trees with subsampling capped at 32. `Numeric AE` (`24-16-8-16-24` on `R0`) uses 100 full-batch Adam epochs ($\text{lr} = 10^{-3}$), avoiding mini-batch noise on 32 rows. **Tier 2 Secondary Models:** `Conditional Residual` fits ridge regression ($\alpha = 10.0$, strong ℓ_2 shrinkage for $p = 16$ predictors on $n_{\text{ref}} = 32$ rows, $p/n = 0.5$) and logistic regression ($C = 0.1$) to predict patch features from workflow features on $\mathcal{R}_{\text{norm}}$, scoring standardized output residuals. `Denoising AE` (`23-16-4-16-23` on `R0-clean`) uses 10% masking, Huber ($\delta = 1.0$) + BCE loss, and 120 AdamW epochs ($\text{lr} = 10^{-3}$, weight decay 10^{-4}). **Tier 3 Supervised References:** `T3` (`R3`) and `R1` (TF-IDF) fit balanced-class ℓ_2 logistic regression ($C \in \{0.01, 0.1, 1, 10\}$); `XGBoost` (Chen and Guestrin, 2016) uses `max_depth=3` (≤ 8 leaves for 170 rows), 100 trees, $\text{lr} = 0.05$, and 0.8 row/col subsampling.

6 Experimental Setup

All models share an identical 5-fold stratified outer split (`seed=42`) on 284 `GPT-4` runs, with inner stratified splits (`seed=100+f`) yielding 60% fitting ($n_{\text{fit}} \approx 170$, $n_{\text{ref}} \in \{32, 33\}$ normal) and 20% validation ($n_{\text{val}} \approx 57$, 10–11 normal). Normal-only models fit strictly on $y = 0$ fitting rows; validation rows select hyperparameters and τ^* without refitting. Test partitions are strictly held out, evaluated once, and never accessed during preprocessing, model fitting, or threshold selection. Small validation normal pools ($n_{\text{val,norm}} \approx 11$) introduce discrete quantile steps in τ^*; pooling and 2,000-resample stratified bootstrap CIs quantify calibra-

Method	Pool BA	Fold BA±SD	FPR	Recall	TN/FP/FN/TP	Fold AP±SD	p-val	Train (s)	ms/run	Holdout BA (ZS / CV)
Always alert (context)	0.500	0.500±0.00	1.000	1.000	0/54/0/230	0.810±0.01	–	–	–	–
Never alert (context)	0.500	0.500±0.00	0.000	0.000	54/0/230/0	0.810±0.01	–	–	–	–
T1: Rule heuristic	0.526	0.526±0.09	0.204	0.257	43/11/171/59	0.864±0.05	0.310	0.00	0.000	0.501 / 0.508
T2: Dual encoder kNN	0.481	0.479±0.05	0.259	0.222	40/14/179/51	0.841±0.02	–	0.69+3151	0.000	0.542 / 0.536
T3: Encoder logistic	0.596	0.594±0.02	0.352	0.543	35/19/105/125	0.905±0.02	0.003	0.00+3151	0.018	0.691 / 0.711
R1: TF-IDF logistic	0.652	0.652±0.05	0.352	**0.657**	35/19/79/151	**0.923**±0.03	0.006	92.06	34.410	–
Inductive ECOD (R0-clean)	0.657	0.654±0.05	**0.130**	0.443	47/7/128/102	0.913±0.03	0.004	0.01	0.003	0.645 / 0.645
Isolation Forest (R0-clean)	0.635	0.632±0.02 (0.015)	0.148	0.417	46/8/134/96	0.914±0.03	0.004	0.26	0.240	0.618 / 0.595
Conditional residual (sec.)	**0.675**	0.671±0.07	0.185	0.535	44/10/107/123	0.908±0.04	0.011	0.01	0.019	0.672 / 0.708
Denoising AE (sec.)	0.648	0.646±0.05 (0.018)	0.148	0.443	46/8/128/102	0.909±0.03	0.020	0.02	2.233	–
Numeric AE (v1 control)	0.651	0.647±0.06 (0.034)	0.185	0.487	44/10/118/112	0.904±0.05	0.002	0.01	0.649	–
XGBoost (v1 control)	0.646	0.643±0.05	0.278	0.570	39/15/99/131	0.908±0.04	0.008	0.02	0.304	0.593 / 0.693

Table 2: Out-of-fold (OOF) operating points over all 284 `GPT-4` trajectories (54 normal, 230 failure) under validation FPR $\leq 20\%$ calibration. **Fold BA±SD**: 5-fold mean±SD (3-seed SD in parentheses for stochastic models). **p-val**: paired t-test across 5 folds vs. `T2`. **Train (s) / ms/run**: per-fold fit time and warmed inference latency (+3151 ms for frozen CPU encoder extraction). **Holdout BA (ZS / CV)**: untouched 271-run `Claude-3-Opus` external holdout BA in Mode A zero-shot cross-agent transfer (**ZS**) and Mode B 5-fold confirmatory replication (**CV**).

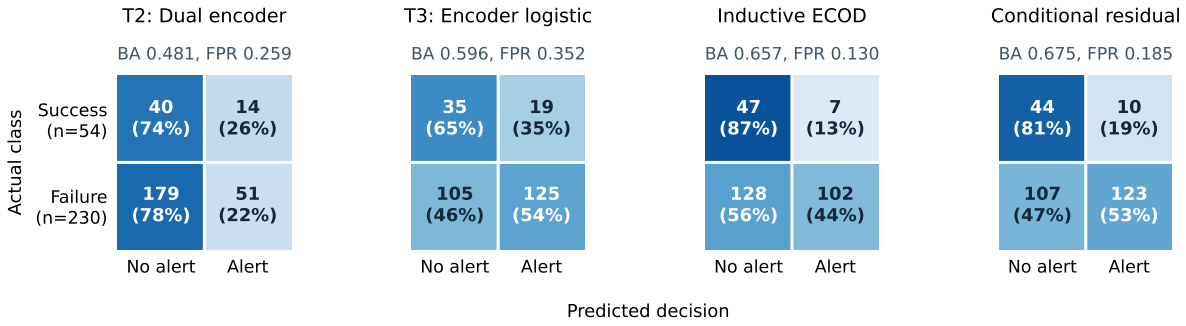


Figure 3: Pooled OOF confusion matrices across 284 `GPT-4` runs ($n_0 = 54$, $n_1 = 230$), colored by row-normalized class percentage. Primary `T2` leaves 219 runs unflagged (40 successes, 179 missed failures; 18.3% success precision).

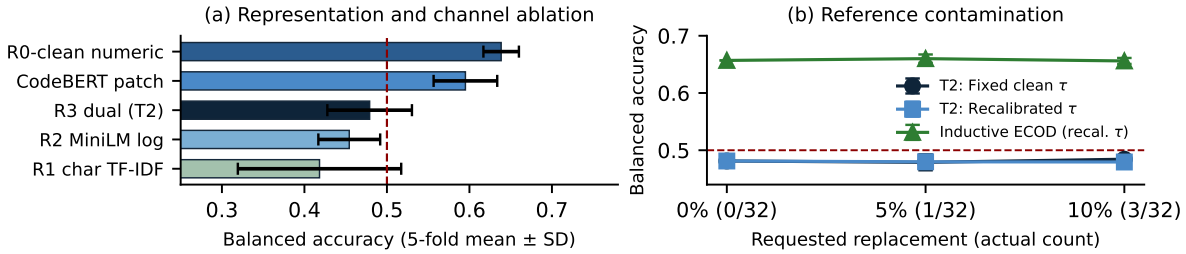


Figure 4: Single-factor ablations: (a) k-NN representation/channel ablation (bars: 5-fold mean±SD), where `R0-clean` (0.638 ± 0.02, pooled OOF 0.641) and `CodeBERT` patch-only (0.595 ± 0.04, pooled OOF 0.594) beat unweighted dual fusion (0.479 ± 0.05, pooled OOF 0.481); (b) AD reference contamination across 0%, 5% (1/32), and 10% (3/32) injected failures for `T2` and `Inductive ECOD`.

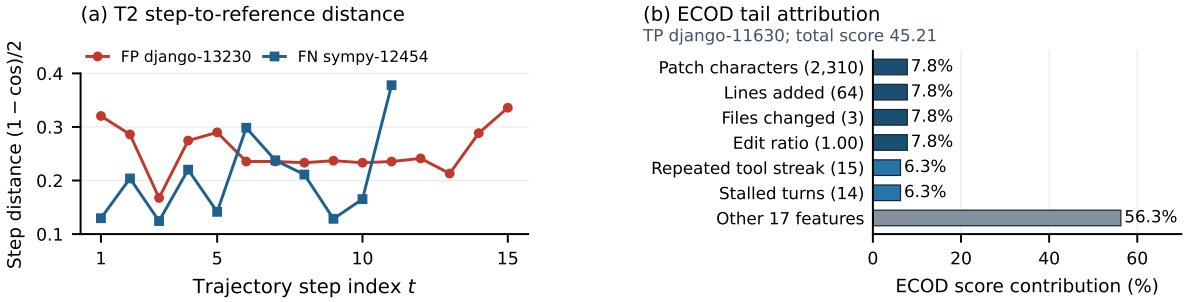


Figure 5: Interpretability and error diagnosis: (a) Step-level `MiniLM` distance $\frac{1-\cos}{2}$ to nearest normal reference for FP `django-13230` ($d_{\log} = 0.203$, noisy debugging before a valid 2-line fix) vs. FN `sympy-12454` ($d_{\log} = 0.047$, quiet run with incomplete logic); (b) Additive per-feature tail attribution for `Inductive ECOD` on TP `django-11630` ($s(x) = 45.21 > \tau^* = 33.42$).

tion uncertainty. Stochastic models run across seeds `42`, `123`, `456`. On the untouched 271-run `Claude-3-Opus` holdout, we run once under two locked protocols: **Mode A (Zero-Shot Trans-**

k-NN Channel	Pooled (Fold±SD)	ΔBA [95% CI]	p	FPR/Rec.
R0-clean num.	**0.641** (0.638±0.02)	+0.159 [+0.09,+0.23]	0.005	0.17/0.45
R1 char TF-IDF	0.419 (0.418±0.10)	-0.062 [-0.15,+0.03]	0.386	0.30/0.13
R2 MiniLM log	0.456 (0.454±0.04)	-0.025 [-0.06,+0.00]	0.160	0.30/0.21
CodeBERT patch	0.594 (0.595±0.04)	+0.113 [+0.03,+0.20]	0.012	0.19/0.37
R3 dual (T2)	0.481 (0.479±0.05)	–	–	0.26/0.22

Table 3: Representation and channel ablation holding k-NN and validation FPR $\leq$ 20% fixed (ΔBA, 95% bootstrap CI, and paired t-test p-value vs. R3 dual T2). Table reports pooled out-of-fold BA with 5-fold mean±SD in parentheses; Figure 4(a) displays fold means.

fer, ZS) applies the 5 frozen GPT-4 fold models and frozen cutoffs τ^* via majority vote, and **Mode B (Confirmatory CV)** executes the locked 5-fold pipeline ($n_{\text{ref}} = 21$ normal fitting runs/fold) without retuning. Because raw accuracy is trivialized by an 81.0% majority-class dummy predictor (BA $= 0.500$), we designate threshold-free **Average Precision** (Fold AP ± SD) as our primary ranking metric across all thresholds, and evaluate discrete triage decisions via **Balanced Accuracy** (BA) under the capacity-constrained FPR $\leq$ 20% threshold τ^*, accompanied by raw confusion counts, 5-fold paired t-tests vs. T2, and 2,000-resample stratified bootstrap 95% CIs.

7 Results

Main three-tier comparison and confirmatory holdout. Table 2 and Figure 3 show primary Tier 2 dual-encoder k-NN (T2, row 4) reaches only BA $= 0.481$ (0.479 ± 0.05), FPR $= 0.259$ ($14/54$), and Recall $= 0.222$ ($51/230$, missing 179 failures). Tier 1 (row 3) reaches BA $= 0.526$ ($p = 0.310$), Tier 3 supervised logistic regression (row 5) reaches BA $= 0.596$ ($p = 0.003$), and R1 TF-IDF (row 6) reaches 0.652 ($p = 0.006$, Fold AP $= 0.923 \pm 0.03$). Crucially, tabular detectors on R0-clean beat all text/code models: Inductive ECOD (row 7) reaches BA $= 0.657$ (ΔBA $= +0.176$, 95% bootstrap CI $[+0.114, +0.240]$, paired t-test $p = 0.004$) with FPR $= 0.130$ ($7/54$, ΔFPR $= -0.130$, CI $[-0.241, -0.037]$), and Conditional Residual (row 9) achieves best overall BA $= 0.675$ (FPR $= 0.185$, Recall $= 0.535$, $p = 0.011$ vs. T2) at 0.003–0.019 ms/run vs. 3,151 ms/run for CPU transformer extraction. On the untouched 271-run Claude-3-Opus holdout (Table 2, last col), Mode A (ZS) and Mode B (CV) exhibit consistent directional rankings: Conditional Residual ($0.672/0.708$, $p = 0.013$), T3 ($0.691/0.711$, $p = 0.013$), and Inductive ECOD ($0.645/0.645$, $p = 0.068$) outperform T2 ($0.542/0.536$) and T1 ($0.501/0.508$). Although 5 folds limit paired t-test power for holdout ECOD ($p = 0.068$), bootstrap CIs confirm the ranking.

Detector (τ mode)	Contam. r/n_{ref}	BA±SD	ΔBA	FPR/Rec.
T2 (Clean τ)	0/32 (0%)	0.481±0.000	+0.000	0.26/0.22
T2 (Recal. τ)	0/32 (0%)	0.481±0.000	+0.000	0.26/0.22
T2 (Clean τ)	1/32 (3.1%)	0.479±0.014	-0.002	0.27/0.22
T2 (Recal. τ)	1/32 (3.1%)	0.480±0.011	-0.001	0.26/0.22
T2 (Clean τ)	3/32 (9.4%)	0.484±0.007	+0.003	0.25/0.22
T2 (Recal. τ)	3/32 (9.4%)	0.479±0.002	-0.002	0.25/0.21
ECOD (Recal. τ)	0/32 (0%)	0.657±0.000	+0.000	0.13/0.44
ECOD (Recal. τ)	1/32 (3.1%)	0.660±0.007	+0.003	0.12/0.44
ECOD (Recal. τ)	3/32 (9.4%)	0.656±0.005	-0.001	0.14/0.45

Table 4: AD-specific normal-reference contamination across 3 seeds (42, 123, 456) at fixed $n_{\text{ref}} \in \{32, 33\}$.

Ablation 1: Representation & channel variance decomposition (H1 & H2). In Table 3 and Figure 4(a), holding k-NN and validation calibration fixed, **H1 is refuted:** R0-clean numeric k-NN reaches 5-fold mean BA $= 0.638 \pm 0.02$ (pooled OOF BA $= 0.641$, ΔBA $= +0.159$, $p = 0.005$), beating all text/code variants. **H2 is supported for patch utility, but equal-weight fusion fails:** CodeBERT patch-only reaches 5-fold mean BA $= 0.595 \pm 0.04$ (pooled OOF BA $= 0.594$, ΔBA $= +0.113$ over R3, $p = 0.012$; $+0.138$ over R2 log-only 0.456; and $0.620/0.592$ on Claude-3-Opus). Variance decomposition of Eq. (1) reveals the exact mechanism: CodeBERT patch distances occupy a narrow anisotropic cone ($d_{\text{patch}} = 0.0092 \pm 0.0051$, $\text{Var}(\frac{1}{2}d_{\text{patch}}) = 6.56 \times 10^{-6}$), whereas MiniLM log distances have $15.6\times$ larger mean and $56.0\times$ larger variance ($d_{\log} = 0.1433 \pm 0.0383$, $\text{Var}(\frac{1}{2}d_{\log}) = 3.67 \times 10^{-4}$). Thus, noisy log distances contribute $\mathbf{98.24}\%$ of unweighted variance in $d(x, r)$, drowning out the patch channel.

Ablation 2: Reference contamination (H3). Table 4 and Figure 4(b) replace $r \in \{0, 1, 3\}$ of 32 normal references (0%, 3.1%, 9.4%) with failures across 3 replacement seeds. **H3 is refuted:** neither T2 ($|\Delta\text{BA}| \leq 0.003$, $p \geq 0.25$) nor Inductive ECOD ($|\Delta\text{BA}| \leq 0.003$, $p \geq 0.34$) degrades monotonically. In R3, contaminated references enter k-NN sets for $9.6\%\pm2.0\%$ ($1/32$) and $31.1\%\pm7.6\%$ ($3/32$) of test runs, but because failure (0.0770 ± 0.0194) and normal (0.0732 ± 0.0200) distances overlap heavily, mean failure scores shift by $<$ 0.01 SD ($+0.00001$ and -0.00014). In ECOD, replacing $\leq$ $3/32$ ranks shifts smoothed tail log-probabilities by at most $\log(33/30) \approx 0.095$ nats/feature, absorbed by validation τ^* recalibration.

Ablation 3: Threshold calibration vs. operational cost model. Table 5 compares threshold rules against our cost model ($C_{\text{Alert}} =$ \$30 vs. $C_{\text{FN}} =$ \$1,200). Under capacity-constrained FPR $\leq 20\%$, T2 achieves recall of only 0.222 (Table 2, row 4; Table 5, row 1), missing 179 failures (\$381,602), while ECOD reaches 0.443 recall, missing 128 failures (\$276,180). Switching to unconstrained validation F_2 / min-cost calibration lowers $\bar{\tau}^*$ to 0.046 and 17.05, raising recall to 0.957

Model (Val Rule)	τ^*	BA	FPR/Rec.	F_2	Cost ($)
T2 (FPR$\leq$20%)	0.091	0.481	0.26/0.22	0.259	$381,602
T2 (Max val BA)	0.070	0.524	0.57/0.62	0.654	$192,993
T2 (Max val F_1)	0.050	0.541	0.85/0.93	0.910	$45,475
T2 (Max F_2/Min$)	0.046	0.515	0.93/0.96	0.924	$35,387
ECOD (FPR$\leq$20%)	33.43	0.657	0.13/0.44	0.496	$276,180
ECOD (Max val BA)	31.08	0.646	0.26/0.55	0.598	$225,053
ECOD (Max val F_1)	17.71	0.540	0.89/0.97	0.936	$29,102
ECOD (Max F_2/Min$)	17.05	0.528	0.93/0.98	0.945	$23,028
Always alert (Bayes)	$-\infty$	0.500	1.00/**1.00**	**0.955**	**$15,000**

Table 5: AD threshold calibration and dollar cost (scaled to 500 runs: $30/review, $1,200/escaped FN; break-even $\pi^* = 2.5\%$).

($F_2 = 0.924$, Table 5, row 4, at FPR $= 0.815$) and 0.983 ($F_2 = 0.945$) and cutting net cost by $> 89\%$ to $35,387 and $23,028. However, Bayes-optimal alerting flags whenever posterior failure probability exceeds $\pi^* = C_{\text{Alert}}/C_{\text{FN}} = 30/1200 = 2.5\%$. Because benchmark failure prevalence is $80.99\% \gg 2.5\%$, **Always alert** ($\tau = -\infty$, $15,000 vs. $365,400 unscreened) beats every automated filter. Detectors add distinct value on the rare 6.0% (17/284) **silent semantic failure** subtype, where `T1` catches $\mathbf{0/17}$ ($\mathbf{0.0}\%$), `ECOD` catches 2/17 (11.8%), `T2` catches 6/17 (35.3%), `T3` catches 7/17 (41.2%), and `Conditional Residual` catches $\mathbf{8/17}$ ($\mathbf{47.1}\%$).

8 Analysis

Traceable error cases, attribution, and analyst trust. Table 6 and Figure 5(a) diagnose three `T2` errors: (1) **High-score FP `django-13230`** (Fold 3, $y = 0$, $s = 0.1128 > \tau^* = 0.1087$): despite 16/32 normal `django` references, 15 steps with 9 tracebacks inflate log distance ($d_{\log} = 0.203$, Figure 5(a)), overriding near-zero diff distance ($d_{\text{patch}} = 0.007$ to `django-11133`); score $s(x) = \frac{1}{2}(0.203 + 0.007) = 0.105$ has $\mathbf{96.7}\%$ log noise; (2) **Low-score FN `sympy-12454`** (Fold 1, $y = 1$, $s = 0.0248 < \tau^* = 0.0731$): an 11-step run (3 cumulative errors, 0 on final step, Table 6) yields a clean 1-line edit ($d_{\log} = 0.047$, $d_{\text{patch}} = 0.003$) missing an index bound; and (3) **Boundary FP `sympy-24213`** (Fold 1, $y = 0$, $s = 0.0733 > \tau^* = 0.0731$, margin $+0.0002$): Fold 1 has only 5/32 normal `sympy` references and 1 normal validation run, causing discrete quantile jumps in τ^* (leave-one-repo-out BA spans 0.25–1.00 across 10 repos).

Shared Blind Spot, Analyst Trust & Concrete Fixes: All three errors reflect *surface-feature overlap between normal and failure manifolds under orderless step pooling*. Exploratory debugging in normal tasks mimics failures, while silent semantic bugs alter code without runtime errors. Engineers distrust alerts triggered purely by noisy tracebacks when diffs look clean (e.g., `django-13230`). Transparent feature attribution (as in `ECOD`, Figure 5(b): patch tails 31.1%, tool loops 12.5%) restores trust by isolating execution noise from diff anomalies, cutting review from 15 to < 2 minutes. Operationally, a false alarm wastes $30 vs. $1,200 for an escaped defect (`sympy-12454`). Architectural fixes: (1) inverse-variance weighting ($w_{\text{patch}} \approx 0.98, w_{\log} \approx 0.02$); (2) repo-stratified calibration τ^*_{repo}; and (3) causal sequence pooling. Cross-agent transfer to `Claude-3-Opus` shifts log tokens by $+18.4\%$; production references must refresh when rolling 50-run FPR $> 25\%$ or upon scaffold upgrades.

Case ID	Type	Fold	St/Er/Ed	$d_{\log}/d_{\text{p}}$	$s(x)$	τ^* (Δ)
`django-13230`	FP	3 (16/32)	15/9/8	0.203/0.007	0.113	0.109 (+0.0041)
`sympy-12454`	FN	1 (5/32)	11/3/1	0.047/0.003	0.025	0.073 (-0.0483)
`sympy-24213`	FP*	1 (5/32)	16/11/1	0.144/0.003	0.073	0.073 (+0.0002)

Table 6: Traceable `T2` error cases (`FP*`: boundary FP) with fold (same-repo normal refs/32), raw **St**eps/**Er**rors (cumulative count; 0 on final step for `sympy-12454`)/**Ed**its, $d_{\log}/d_{\text{patch}}$, score $s(x)$, and τ^* ($s - \tau^*$).

9 Conclusion and Limitations

Across 284 `GPT-4` runs and an untouched 271-run `Claude-3-Opus` holdout, frozen dual-encoder k-NN (`T2`, BA $= 0.481$) underperforms parameter-free `Inductive ECOD` (BA $= 0.657$, $p = 0.004$) and `Conditional Residual` (BA $= 0.675$, $p = 0.011$). Unweighted fusion fails because noisy log distances carry 98.24% of channel variance, drowning out `CodeBERT` diff embeddings (BA $= 0.594$, $p = 0.012$). Under release costs ($\pi^* = 2.5\%$), 81% failure prevalence favors universal review ($15,000) for general triage, whereas detectors uniquely catch 47.1% of rare (6.0%) silent semantic failures missed by surface rules (0.0%). Limitations include small normal pools ($n_{\text{ref}} \in \{21, 32\}$), 10-normal validation quantization, orderless pooling, and execution-free diff encoding; deployment requires recalibration when rolling 50-run FPR exceeds 25% or upon scaffold upgrades.

Contributions of Each Member.

- Tun-Chieh Yang (315553010): Pending confirmation.
- Jia-Cheng Wu (314554059): Pending confirmation.
- Min-Quan Wang (315553008): Pending confirmation.
- Wei-Chia Chang (315553011): Pending confirmation.
- Chun-Ting Hou (315554022): Pending confirmation.

AI Usage Statement. OpenAI Codex and Google Antigravity assisted with pipeline implementation, local embedding extraction, holdout evaluation, and formatting. Numerical claims are verified via `tools/verify_submission_v2.py`; contributions remain pending team confirmation.

References

Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.

Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, Ming Gong, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, and Ming Zhou. 2020. Codebert: A pre-trained model for programming and natural languages. In *Findings of EMNLP*, pages 1536–1547.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations*.

Zheng Li, Yue Zhao, Xiyang Hu, Nicola Botta, Cezar Ionescu, and George H. Chen. 2022. Ecod: Unsupervised outlier detection using empirical cumulative distribution functions. *arXiv preprint arXiv:2201.00382*.

Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. 2008. Isolation forest. In *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining*, pages 413–422.

Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992.

Lukas Ruff, Robert Vandermeulen, Nico Goernitz, Lucas Deecke, Shoaib Ahmed Siddiqui, Alexander Binder, Emmanuel Müller, and Marius Kloft. 2018. Deep one-class classification. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80, pages 4393–4402. PMLR.

John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. 2024. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems*, volume 37.